

Rank	Target function	Size	Circuit score	Growth score	Combined score	Part names of circuit components
1	0x33	5-reg	2,511.68	0.843	1.6062	BetI/E1 → BM3R1/B3 → QacR/Q2 → SrpR/S4 → PhIF/P2
2	0x0F	5-reg	2,511.68	0.843	1.6062	BetI/E1 → BM3R1/B3 → QacR/Q2 → SrpR/S4 → PhIF/P2
3	0xCC	6-reg	3,386.61	0.814	1.5862	BM3R1/B3 → QacR/Q2 → AmtR/A1 → BetI/E1 → SrpR/S4 → PhIF/P2
4	0x03	5-reg	1,016.76	0.852	1.5065	SrpR/S4 → BM3R1/B3 → BetI/E1 → QacR/Q2 → PhIF/P2
5	0xEE	4-reg	383.87	0.902	1.4817	BetI/E1 → BM3R1/B2 → QacR/Q2 → PhIF/P2
6	0x07	5-reg	650.27	0.819	1.3840	BM3R1/B2 → SrpR/S4 → BetI/E1 → QacR/Q2 → PhIF/P2
7	0x06	6-reg	469.91	0.821	1.3455	BM3R1/B2 → SrpR/S4 → AmtR/A1 → BetI/E1 → QacR/Q2 → PhIF/P2
8	0x66	6-reg	514.83	0.787	1.2886	BM3R1/B2 → AmtR/A1 → BetI/E1 → SrpR/S4 → QacR/Q2 → PhIF/P2
9	0x2B	6-reg	510.21	0.781	1.2763	AmtR/A1 → BetI/E1 → SrpR/S4 → BM3R1/B3 → QacR/Q2 → PhIF/P2
10	0x8E	5-reg	328.42	0.800	1.2570	BM3R1/B3 → SrpR/S4 → BetI/E1 → QacR/Q2 → PhIF/P2
11	0x3C	6-reg	432.37	0.775	1.2429	BM3R1/B2 → BetI/E1 → AmtR/A1 → QacR/Q2 → SrpR/S4 → PhIF/P2
12	0x77	4-reg	73.20	0.874	1.2122	BM3R1/B2 → BetI/E1 → QacR/Q2 → PhIF/P2
13	0xE8	6-reg	169.37	0.801	1.1748	BM3R1/B2 → AmtR/A1 → SrpR/S4 → BetI/E1 → QacR/Q2 → PhIF/P2
14	0x4D	5-reg	65.41	0.847	1.1431	SrpR/S2 → QacR/Q2 → AmtR/A1 → BM3R1/B2 → PhIF/P2
15	0x71	5-reg	65.41	0.847	1.1431	SrpR/S2 → QacR/Q2 → AmtR/A1 → BM3R1/B2 → PhIF/P2

Rank	Part	Table S1	Table S2	Total
1	PhIF/P2	15	15	30
2	QacR/Q2	10	15	25
3	BetI/E1	10	13	23
4	AmtR/A1	11	8	19
5	SrpR/S4	8	11	19